

MICROPROCESSORS AND MICROCONTROLLERS LAB

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EXPT. NO. 1	NAME MAANAV R VORA 22BCE3957	M T W T F S S	
		Page No.:	
		Date:	YOUVA

TASK - 1

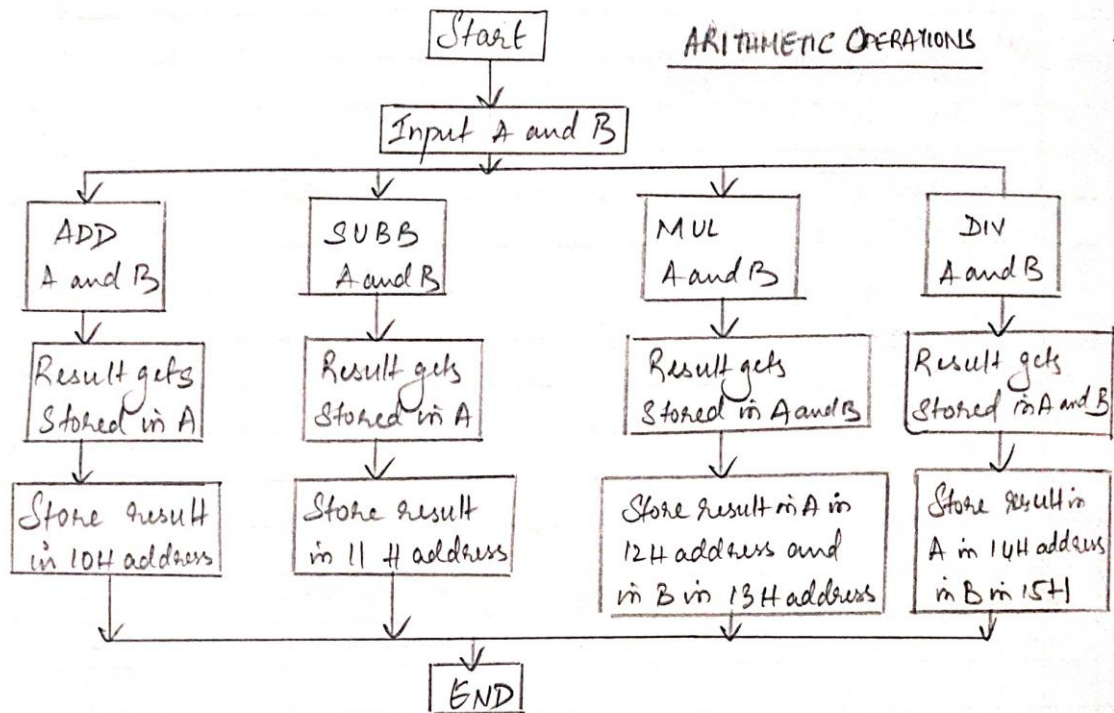
AIM: To perform various basic operations such as addition, subtraction, division, multiplication, AND, OR, XOR, complement, clear in 8 bit hexadecimal numbers in 8051 ALP.

ALGORITHM:

- 1) Initialize A to 25H and B to 12H.
- 2) Add the value in register A and B. The result is stored in A.
- 3) Move the result from register A to 10H address.
- 4) Reinitialize A to 25H
- 5) Subtract the value in register A and B. The result is stored in B.
- 6) Move the result from register A to 11H address.
- 7) Reinitialize A to 25H
- 8) Multiply the value in register A and B.
- 9) Move the result from register A to 12H address and register B to 13H address.
- 10) Reinitialize A to 25H and B to 12H
- 11) Divide the value in register A from B.
- 12) Move the result from register A to 14H address and register B to 15H address.
- 13) Reinitialize A to 25H and B to 12H.
- 14) AND the value in register A and B.
- 15) Move the result from A to 16H address, and reinitialize A as 25H and B as 12H.
- 16) OR the value in register A and B.
- 17) Move the result from A to 17H address, and reinitialize A as 25H and B as 12H.
- 18) XOR the value in register A and B.
- 19) Move the result from A to 18H address, and reinitialize A as 25H and B as 12H
- 20) Complement A register's value and store it in 19H address.
- 21) Clear A, and move its value to 20H address.
- 22) End the program.

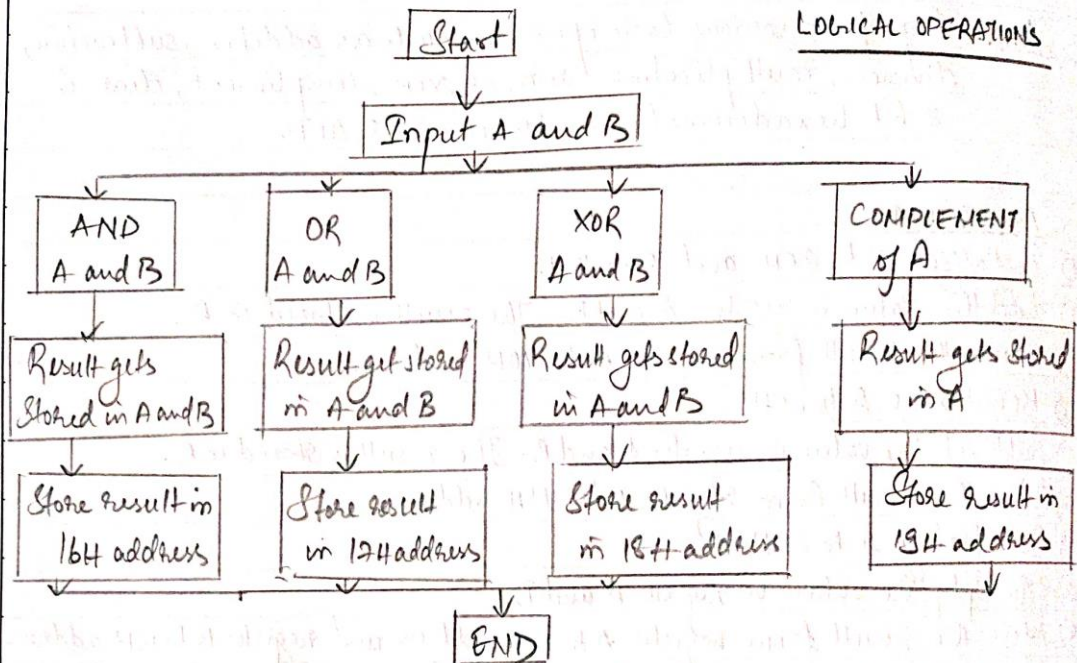
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FLOWCHART



Flowchart

LOGICAL OPERATIONS



EXPT. NO.	NAME	M T W T F S S						
		Page No.:		YOUVA				
		Date:						
	CODE:							
	// ADDITION							
	ORG 0000H							
	MOV A, #25H							
	MOV B, #12H							
	ADD A, B							
	MOV 10H, A							
	// SUBTRACTION							
	MOV A, #25H							
	SUBB A, B							
	MOV 11H, A							
	// MULTIPLICATION							
	MOV A, #25H							
	MUL AB							
	MOV 12H, A							
	MOV 13H, B							
	// DIVISION							
	MOV A, #25H							
	MOV B, #12H							
	DIV AB							
	MOV 14H, A							
	MOV 15H, B							
	// AND							
	MOV A, #25H							
	MOV B, #12H							
	ANL A, B							
	MOV 16H, A							

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EXPT. NO.	NAME	M	T	W	T	F	S	S
		Page No.:						YOUVA
		Date:						
	//OR							
	MOV A, #25H							
	MOV B, #12H							
	ORL A, B							
	MOV 17H, A							
	//XOR							
	MOV A, #25H							
	MOV B, #12H							
	XRL A, B							
	MOV 18H, A							
	//COMPLEMENT							
	CPL A							
	MOV 19H, A							
	//CLEAR							
	CLR A							
	MOV 20H, A							
	HALT: SJMP HALT							
	END							
	<u>CALCULATION:</u>							
	//add				// and			
	A (#25H) : 00100101				A (#25H) : 00100101			
	B (#12H) : 00010010				B (#12H) : 00010010			
	<u>00110111</u>				<u>00000000</u>			

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EXPT. NO.	NAME	M T W T F S S	Page No.	YOUVA
			Date	

subtraction

// OR

A (#25H) 00100101

A (#25H) 00100101

B (#12H) 00010010

B (#12H) 00010010

00010011

00110111

// Multiplication

// XOR

A (#25H) 00100101

A (#25H) 00100101

B (#12H) 00010010

B (#12H) 00010010

100101100

00110111

// Complement

A (#25H) 00100101

11011010

Result:

Values are stored in A and B, ^{are} moved to the respective memory addresses, and hence arithmetic and logical operations are carried out.

22/11/24

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Code:

The screenshot shows the Keil IDE interface. On the left, the 'Registers' window is open, displaying a list of registers under 'Regs' and 'Sys' categories. The 'b' register is highlighted in blue, showing a value of 0x12. Other registers like r0-r7, a, sp, and PC are also visible. On the right, the main assembly editor shows the file 'mv_1.asm'. The code includes instructions such as ORG 0000H, ADD, MOV A, #25H, MOV B, #12H, ADD A, B, MOV 10H, A, SUB A, B, MOV 11H, A, MUL AB, MOV A, #25H, MOV 12H, A, MOV 13H, B, DIV A, B, MOV 14H, A, MOV 15H, B, AND, MOV A, #25H, MOV B, #12H, ANL A, B, MOV 16H, A, OR, and MOV A, #25H. The code is color-coded for readability.

The screenshot shows the Keil IDE interface. On the left, the 'Registers' window displays a list of registers. The 'Sys' section is expanded, and the 'b' register is highlighted with a blue background, showing a value of 0x12. Other registers like r0-r7 are also visible with values of 0x00. On the right, the 'mv_1.asm' assembly file is open, showing a list of assembly instructions. The instructions are color-coded: green for registers, red for immediate values, and blue for instructions. The instructions include DIV, MOV, AND, ANL, OR, XOR, CPL, CLR, and HALT. The assembly code is as follows:

```

19 DIV AB
20 MOV 14H, A
21 MOV 15H, B
22 //AND
23 MOV A, #25H
24 MOV B, #12H
25 ANL A, B
26 MOV 16H, A
27 //OR
28 MOV A, #25H
29 MOV B, #12H
30 ORL A, B
31 MOV 17H, A
32 //XOR
33 MOV A, #25H
34 MOV B, #12H
35 XRL A, B
36 MOV 18H, A
37 //COMP
38 CPL A
39 MOV 19H, A
40 //CLEAR
41 CLR A
42 MOV 20H, A
43
44 HALT: SJMP HALT
45 END

```

Output:

[illegible]